

Y

 $\mathbf{p}_k(x_k, y_k)$

Path segment

 θ_r (x, y) y_{robot} e_y e_θ

$$\mathbf{d} = \begin{bmatrix} x_r - x \\ y_r - y \end{bmatrix}$$

 e_x x_{robot} θ $\mathbf{p}_{k+1}(x_{k+1}, y_{k+1})$
 $(x_r, y_r, v_r, \omega_r)$

X

